



INTERNATIONAL MATHEMATICS SUMMER CAMP IMSC25

PERFECT SQUARES

VLAD MATEI

1. Find all positive integers n such that $4n! - 4n + 1$ is a perfect square.
2. Let k be a positive real number such that $\lfloor kn^2 \rfloor$ is perfect square for all $n \in \mathbb{N}$. Show that k must be a perfect square.
3. Find all positive integers n such that $2^n - n^2 + 1$ is a perfect square.
4. Given positive integer a and prime number $p \geq 5$. Prove that $2 + a + a^2 + \dots + a^{p-1}$ can't be a perfect square.
5. Are there infinitely many n such that $d(n) \cdot \sigma(n)$ is a perfect square?
6. Find all positive integers k for which there is an irrational $\alpha > 1$ and a positive integer N such that $\lfloor \alpha^n \rfloor$ is a perfect square minus k for every integer n with $n > N$.
7. Show that if x, y, z are positive integers, then $(xy + 1)(yz + 1)(zx + 1)$ is a perfect square if and only if $xy + 1, yz + 1, zx + 1$ are all perfect squares.
8. Suppose that a, b are two natural numbers such that $(a^n + n)(b^n + n)$ is perfect square for all $n \in \mathbb{N}$. Show that $a = b$.
9. Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that $af(a + b) + bf(a) + b^2$ is a perfect square for any pair of natural numbers a and b .
10. Let $a_1, a_2, \dots, a_{2^{2016}}$ be positive integers not bigger than 2016. We know that for each $n \leq 2^{2016}$, $a_1 a_2 \dots a_n + 1$ is a perfect square. Prove that for some i , $a_i = 1$.
11. Show that $n^7 + 7$ is never a perfect square.
12. Find all n such that $(2^n - 1)(5^n - 1)$ is a perfect square.
13. x and y are positive integers and $y(x + 1), x(y + 1)$ are perfect squares. Show that exactly one of x, y is a perfect square.
14. Given an integer $n \geq 4$. $S = \{1, 2, \dots, n\}$. A, B are two subsets of S such that for every pair of $(a, b), a \in A, b \in B, ab + 1$ is a perfect square. Prove that
$$\min\{|A|, |B|\} \leq \log_2 n.$$
15. Find all odd integers n such that $\frac{2^{\phi(n)} - 1}{n}$ is a perfect square.

16. Find all positive integers such that $a + b$ and $ab - 1$ are perfect square at the same time.

17. Find infinitely many triples (a, b, c) of positive integers such that a, b, c are in arithmetic progression and such that $ab + 1$, $bc + 1$, and $ca + 1$ are perfect squares.

18. Let a and b be positive integers, whose difference is a prime number. Prove that $(a^n + a + 1)(b^n + b + 1)$ is not a perfect square for infinitely many positive integers n .

19. Find all primes p such that $p^p - 2$ is a perfect square.